

WATER LEVEL And TEMPERATURE MONITOR

With TFT Graphics Using Arduino

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Water and temperature monitoring systems are crucial across various industries, including the chemical, pharmaceutical, and textile sectors. They help maintain precise liquid levels and stable temperatures—factors essential to ensuring product

quality. In chemical processing, for instance, both parameters must remain within defined limits throughout the operation.

This system uses contact or non-contact water level sensors along with a probe-type temperature sensor. A digital display provides

continuous, real-time monitoring of liquid level and temperature.

Traditional systems often rely on basic LED indicators or 7-segment displays, which do not show actual water volume (in litres) or precise temperature (in degrees Celsius). Some modern systems do feature LCDs that display both parameters with numerical accuracy.

This system demonstrates water

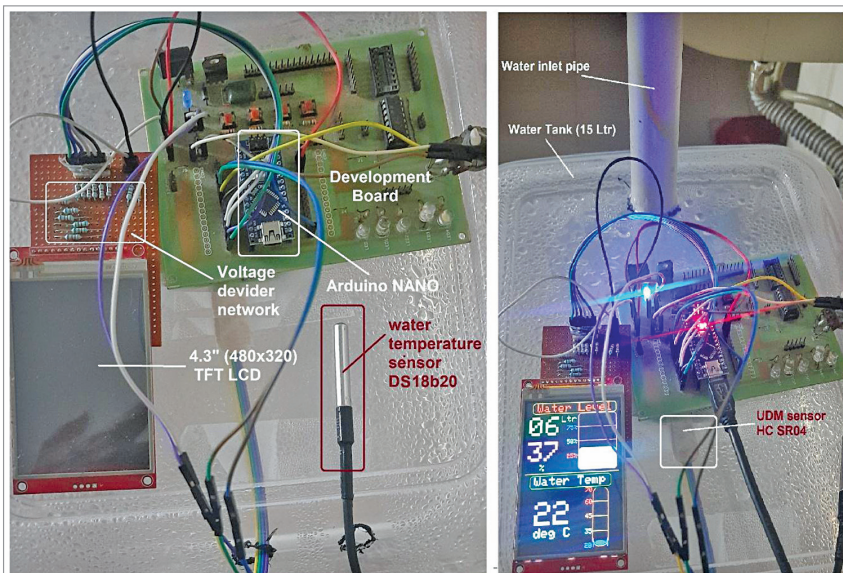


Fig. 1: Author's prototype (L) and a snapshot of the TFT display (R)

TABLE 1
BILL OF MATERIALS

Components	Quantity
Arduino Nano (MOD1)	1
Temperature sensor DS18B20 (S1)	1
HC-SR04 ultrasonic sensor (S2)	1
10.92cm (4.3-inch) SPI TFT display (MOD2)	1
Resistors 1kΩ (R1-R6)	6
Resistors 2.2kΩ (R7-R12)	6
Resistor 4.7kΩ (R13)	1
USB connector	1

Note: The resistors are used as voltage level shifters. Some displays support multi-voltage logic (3.3V and 5V), in which case logic level shifter resistors are not needed.

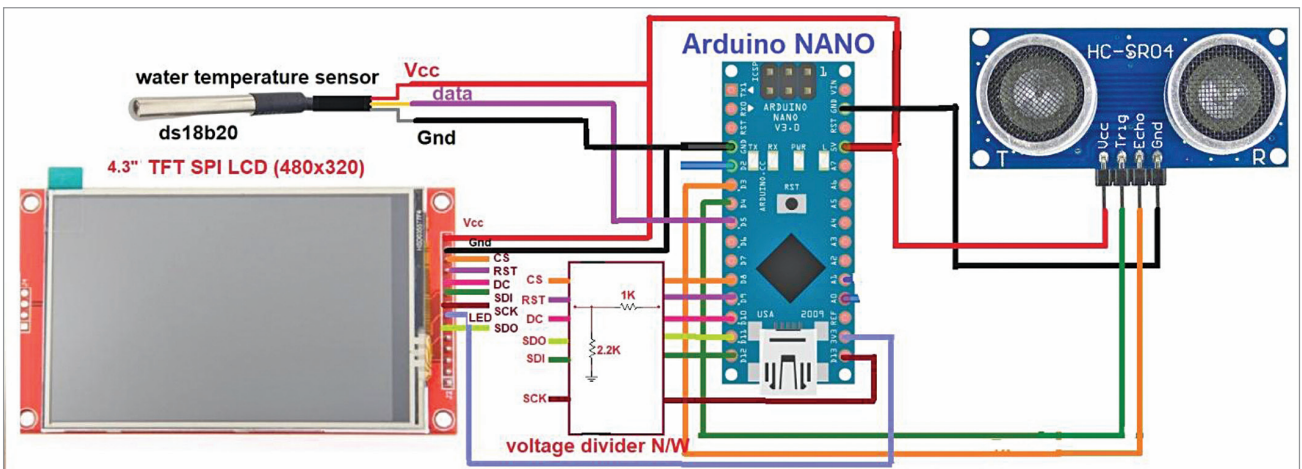


Fig. 2: Components connection